
Task-Aware Mechanism: Hybrid MoE Vision Tower

Towards Holistic Video Understanding

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Abstract

Should an LVLM use the same video processing pipeline for *comprehending the main idea of a 2-hour movie* and *counting the birds appearing in a 15-second clip*? We present Task-Aware Mechanism (TAM), a hybrid-gated Mixture-of-Experts (MoE) vision tower that adapts frame count and resolution to the user query and video length. TAM introduces a compact 0.1B text-only *Inductor* trained on our TA-116K dataset to infer task types, enabling task-aware visual budgeting and routing: a soft-gated MoE vision encoder for stability, and hard-gated resolution-specific projectors/pipelines for efficient specialization. The explicit budget also separates task classification from downstream visual-language reasoning, making routing decisions easier to inspect and helping diagnose when fixed visual pipelines under-sample temporal or spatial evidence. Built on Qwen2-7B, TallVA-8B-A7B achieves state-of-the-art performance among models with comparable LLMs on most video benchmarks and remains competitive against stronger-LLM baselines, showing that task-aware visual budgeting makes video understanding more holistic.

1. Introduction

The recent breakthroughs in large language models (LLMs) (OpenAI, 2023; DeepSeek-AI, 2024) have sparked the emergence of large vision-language models (LVLMs) (Zhang et al., 2025c; Yang et al., 2023; Bai et al., 2025; Hong et al., 2024), which integrate visual and linguistic capabilities for vision-centric multimodal understanding. Influenced by LLM scaling laws (Kaplan et al., 2020), traditional LVLM

paradigms have often focused on improving performance by adding more parameters. However, larger single-model LVLMs often consume multiples of the computational resources required by smaller counterparts while yielding only marginal performance gains. The success of Mixture-of-Experts (MoE) architectures has challenged the conventional preference for monolithic models. More crucially, MoE has reignited interest in task-aware design for video understanding (Li et al., 2024c; Ranasinghe et al., 2025; Tan et al., 2024; Ataallah et al., 2024).

We define task awareness as the ability, within a unified model architecture, to perceive distinct task characteristics, dynamically select appropriate tools, experts, or processing pipelines, and achieve better performance with fewer or comparable activated parameters. Interpretable MoE architectures, frame-selection methods (Ranasinghe et al., 2025; Tan et al., 2024), and token-selection strategies all embody this task-aware principle from different angles.

Current LVLMs have widely adopted sparse MoE architecture for LLMs (MoE-LLM) (Zhang et al., 2025a; Lin et al., 2024a), which has proven effective for scaling language capacity. However, research on the Vision Tower (VT) remains comparatively limited. As the "eye of LVLMs", the VT, typically composed of a Vision Encoder (VE), frame processing pipelines, and a Projector, critically determines the quality of visual information available to the LLM. A few attempts (Zhang et al., 2025a; Riquelme et al., 2021; Li et al., 2024a) have replicated MoE-style scaling inside vision components, yielding only marginal gains and indicating that merely expanding capacity is insufficient.

The main issue lies in the rigid visual processing paradigm of conventional VTs: all videos are uniformly converted into frame sequences with a fixed count and resolution, as shown in Fig. 1(a). This paradigm neglects the decisive role of the user query in defining task requirements. For the same video, different queries may demand entirely different visual information: some require high-resolution frames for fine-grained recognition, while others need dense temporal coverage to follow dynamic events over time.

This analysis reveals that advancing LVLMs requires a task-aware VT. Such a VT should adaptively decide video pro-

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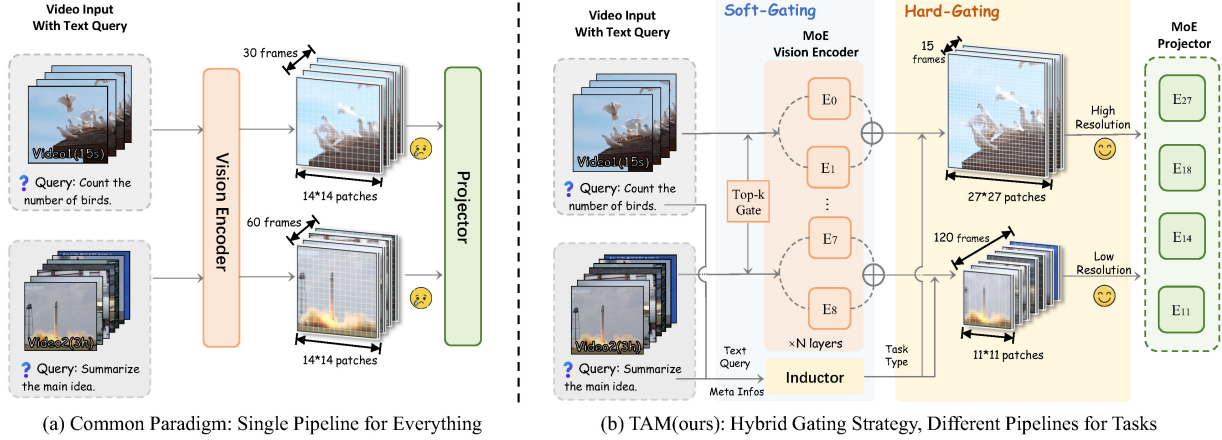


Figure 1. Comparison of common paradigms (a) and TAM (b). (a) Common paradigms use a single pipeline for all tasks, such as max_fps=2 and max_frame=60. (b) TAM uses the Inductor to perceive video length and user query, then assigns appropriate frame counts and resolutions for different tasks.

cessing strategies based on the user query’s intent, providing appropriate visual frames to the LLM. The MoE paradigm, which routes inputs to specialized experts, aligns naturally with this task-aware perspective. To design an efficient and suitably gated MoE mechanism for different VT components, there are two critical challenges:

(1) Lack of Datasets. To the best of our knowledge, no existing dataset jointly annotates query sensitivity to both frame count and spatial resolution for visual budget routing in video LVLMs. Existing benchmarks may include duration metadata or question-type labels, but they do not directly specify the task-specific visual requirements TAM needs for reliable routing across videos.

(2) Lack of Adaptive Gating Strategies. Since VT contains both continuous neural modules and discrete video-processing stages, a traditional soft-gating MoE strategy cannot be uniformly applied across all components. Soft-gating, as shown in Fig. 3(b), improves the stability of VE routing, whereas hard-gating, as illustrated in Fig. 3(a), is more suitable for deterministic choices such as frame count, pooled resolution, and projector expert. Employing either strategy in isolation appears to be suboptimal across diverse tasks and video lengths within one VT.

To address these challenges, we propose Task-Aware Mechanism (TAM). TAM has three primary components: a task classification framework and the TA-116K query-label dataset, a 0.1B text-only Inductor module that predicts task types from the user query and video length, and a hybrid-gated MoE-VT that translates those predictions into task-specific frame counts, pooled resolutions, and projector routes. We summarize our primary contributions as follows:

- We introduce TAM, a Hybrid Sparse Gated MoE-VT architecture (Fig. 1(b)). We show that MoE-ViT and Dynamic Frame Number & Resolution (DFR) work best together:

neither component alone explains the gains, while their combination is strongest under fixed compute and context budgets.

- We categorize video understanding tasks according to their sensitivity to frame count and resolution, and construct TA-116K with 116K annotated queries from selected video datasets for future benchmarks and model training.
- Using TAM, we train TallVA-8B-A7B (8B total / 7B active), which achieves the best result on most benchmarks among comparable-LLM models and remains competitive against stronger-LLM baselines (Tab. 1, Fig. 6).

The appendix provides qualitative cases illustrating how visual-budget choices affect model behavior and can guide low-cost routing work for video LVLMs.

2. Related Work

Single LVLMs. The evolution of LVLMs builds on vision-language alignment frameworks such as Flamingo (Alayrac et al., 2022) and BLIP-2 (Li et al., 2023), with the LLaVA series (Liu et al., 2023; 2024b;c; Zhang et al., 2024; Li et al., 2025) popularizing visual instruction tuning. Scaling efforts span stronger vision encoders (Bai et al., 2023; Chen et al., 2024c; Lin et al., 2023; Liu et al., 2024a), connector optimization (Cha et al., 2024; Lin et al., 2024b), and larger multimodal corpora (McKinzie et al., 2024; Li et al., 2024d). Video extensions (Li et al., 2024c) tackle temporal modeling through token compression, but still face context window limitations. Recent works further improve visual token efficiency via dynamic token merging (Jin et al., 2024), instruction-guided visual token pruning (Huang et al., 2024), spectrum-preserving token merging (Tran et al., 2024), and adaptive positional encoding (Zeng et al., 2025).

MoE-LVLMs. Mixture-of-Experts (MoE) architectures enhance LVLMs by conditional computation, as seen in

language models (Fedus et al., 2022; Dai et al., 2024). MoE-LLaVA (Lin et al., 2024a) demonstrates the efficacy of sparse computation for visual reasoning, while DeepSeek-VL2 (Wu et al., 2024b) achieves modality-specific expert specialization. Prior vision-side MoE studies such as CLIP-MoE (Zhang et al., 2025a), V-MoE (Riquelme et al., 2021), and CuMo (Li et al., 2024a) mainly improve capacity or reuse pretrained experts inside vision components, but they do not jointly route the video-processing budget according to the user query. DynFocus (Han et al., 2025) optimizes spatiotemporal routing. Key challenges include load balancing and expert routing, studied through variance regularization (Zoph et al., 2022), multimodal contrastive learning (Mustafa et al., 2022), fine-grained sparse routing in DBRX (Gupta & Yip, 2024), and adaptive vision MoE design (Chen et al., 2023).

LVLMS with Task-Aware Insights. Task-aware design has been gaining attention with the application of MoE architectures, and its core idea is "right tool for the right job": different tasks should use different processing methods. Early video methods used uniform sampling (Maaz et al., 2024b) or relevance scoring (Yu et al., 2023) for frame selection, while modern approaches include dynamic token compression (Li et al., 2024c), lightweight frame selection (Ranasinghe et al., 2025; Tan et al., 2024), query-aware frame querying (Yu et al., 2025), M-LLM-based frame selection (Hu et al., 2025), hierarchical processing (Azad et al., 2025), and memory-augmented retrieval (Ataallah et al., 2024). These methods primarily select or compress frames for a monolithic vision tower. TAM instead couples query- and duration-aware frame/resolution budgeting with a hybrid MoE-VT, so the selected visual budget also determines the projector route and supports expert specialization. A related practice (Wu et al., 2025) uses multiple Vision Encoders with different specifications to collect richer visual cues such as depth and color; TAM is complementary, focusing on task-aware temporal and spatial budgeting inside a unified video LVLMS interface. Additional recent related work is discussed in the appendix.

3. Methodology

3.1. Preliminary

MoE architecture. The router assigns tokens to experts and calculates the weight matrix $W \in \mathbb{R}^{N \times M}$, where N and M represent the number of tokens and experts, respectively. In the Dense MoE method, each token is assigned to all experts, and the output O is computed as:

$$O_i = \sum_{j=1}^M W_{i,j} E_j(I_i), \quad O \in \mathbb{R}^{N \times D_{out}} \quad (1)$$

Here, E_j denotes the j -th expert, D denotes the dimension, and $I \in \mathbb{R}^{N \times D_{in}}$ is the input. To reduce computational costs,

the Sparse MoE method assigns each token to only the top- K experts with the highest weights. The recalculated weight matrix is:

$$W' = \text{Softmax}(\text{TopK}(W)), \quad W' \in \mathbb{R}^{N \times M} \quad (2)$$

Here, $\text{TopK}(W)$ retains only the top- K elements in each row of W , setting all other elements to zero.

Weight Initialization and Gating Strategy. Conventional MoE models initialize their experts with unique weights, while Co-upcycling (He et al., 2024b; Komatsuzaki et al., 2023) initializes all experts with the same weights derived from a pre-trained checkpoint. Soft-gating employs a trainable network to compute W , offering high performance but increasing training complexity; in contrast, hard-gating uses algorithms to compute W , simplifying training and facilitating hypothesis validation (Fig. 3).

3.2. Task-Aware Mechanism

Classification strategy, Fig. 2(a). We collected 10k representative queries from various video understanding datasets (Zhang et al., 2025c; Chen et al., 2024b; Farré et al., 2024; Rawal et al., 2024; Share, 2024; Maaz et al., 2024a). We classify the tasks into 8 visual-budget categories according to their intrinsic meaning and, most importantly, their sensitivity to resolution and frame number. These categories are not a mutually exclusive semantic ontology; instead, they define the dominant visual-budget regime used by TAM. For example, Temporal Sequence Positioning is much more sensitive to the number of frames than resolution, while Text Recognition (OCR) prefers higher resolution. We further examine this routing abstraction through a coarsening ablation, with task definitions and prompts provided in the appendix.

Inductor and Dynamic Frame Selection, Fig. 2(b). As outlined in the Introduction, to develop a more robust router capable of considering both query and video length, we designed the Inductor. Specifically, we chose a lightweight pre-trained text model, SmoLLM2-135M-Instruct (Allal et al., 2025), as the base model for the Inductor. The video duration is serialized into the text prompt as both seconds and a duration bin, together with the user query. The last layer of the Inductor is not the typical lm-head for text output, but a sequence classification layer that outputs a softmax probability array over the 8 task categories.

In previous paradigms, all tasks shared the same frame processing pipeline. Videos are first subjected to uniform frame sampling and fed into the VE, followed by spatial pooling of the patch outputs (e.g., from 27×27 to 14×14), and finally to the LLM through the Projector. In TAM, the information flow can be summarized as: Query + video metadata $\rightarrow$ Inductor $\rightarrow$ DFR scheduler $\rightarrow$ frame sampling

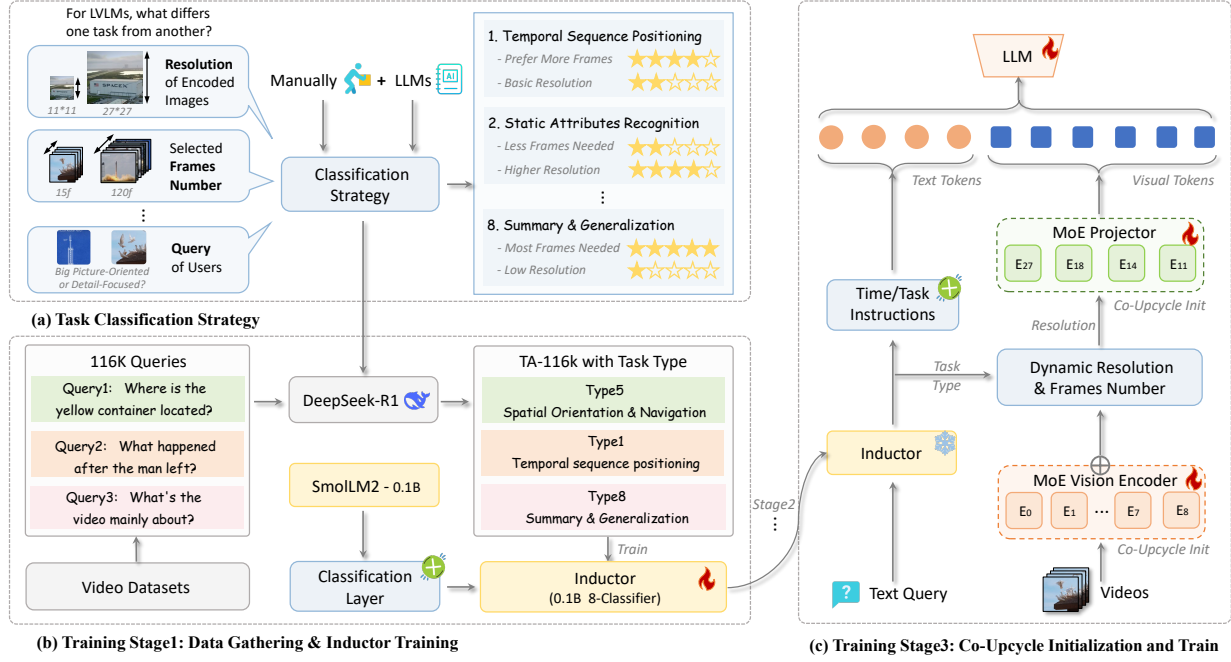


Figure 2. Overview of our work. (a) Considering video metadata and user queries, we organize about 10k representative queries into 8 visual-budget categories according to their different sensitivities to resolution and frame number. (b) In training Stage 1, we use the classification strategy to organize 116K queries, label them with DeepSeek-R1, and train the Inductor. (c) In training Stage 3, after the LLM has adapted to Dynamic Frame Number and Resolution in Stage 2, we initialize the VE and Projector in MoE-VT by Co-Uncycling and train the final TallVA model used in our experiments.

and pooled token-grid selection $\rightarrow$ MoE-VE $\rightarrow$ resolution-specific Projector $\rightarrow$ LLM. Specifically, under a fixed visual-token budget, the upper bound for frame count is set to 120 following previous works, and the pooled per-frame token grid is selected from $R \times R$, where $R \in \{27, 18, 14, 11\}$ ¹. The scheduler is deterministic: it uses task-specific resolution priorities, duration thresholds of 30s and 180s, and minimum sampling intervals of 0.5s, 1s, and 2s for short, medium, and long videos, respectively, then selects the first feasible frame-count/resolution pair satisfying the token budget. This rule-based design makes routing auditable and keeps the Inductor responsible for task-regime prediction rather than low-level frame optimization; the appendix lists representative scheduler outputs.

Hybrid Sparse Gating Strategy. Fig. 2(c). We apply the Sparse Soft-Gating MoE Strategy for the VE. Given that the VE uses a Transformer architecture, the Soft-Gated design effectively enhances stability when handling complex tasks. For the Projector, we employ four Hard-Gated resolution-specific experts corresponding to the 27×27 , 18×18 , 14×14 , and 11×11 pooled token grids. This Hybrid Gating Strategy is also shown in Fig. 1(b). Through ablation

¹ SigLIP-so400m-patch14 produces a default 27×27 patch grid per frame; $\{27, 18, 14, 11\}$ are the rounded pooled grid sizes obtained with different stride factors, respectively. They are not fixed hyperparameters for all models.

studies in Sec. 4.3 and Tab. 3, we demonstrate that our Hybrid Gating Strategy maintains low activated parameters while respecting the distinct characteristics of the VE and Projector, enhancing overall performance.

3.3. Three-Stage Training Recipes

Data Collecting and Inductor Training. Stage 1 (Fig. 2(b)) trains the Inductor on TA-116K. We select 116K queries from video datasets (Li et al., 2025; Zhang et al., 2025c), then annotate them with our classification strategy and DeepSeek-R1 (Guo et al., 2025). Each sample contains serialized video-duration metadata, a text query, and a classification label. Fig. 4(a) shows the data distribution; detailed annotation prompts and training settings are deferred to the appendix.

LLM Adaptation to DFR. In Stage 2, we empirically find that the LLM should be trained prior to the Projector in TAM, contrary to the classical paradigm (Li et al., 2025; Liu et al., 2023; 2024b;c) where Projector training is conducted first to align the vision side with the LLM. After multiple unsuccessful attempts using standard methodologies, we achieved improved results by prioritizing LLM training at this stage. We analyze this training order in Sec. 4.3.

MoE Initialization and Training. In Stage 3 (Fig. 2(c)), we train all components of TallVA except the Inductor, which



Figure 3. Different gating strategies. (a) Hard gating strategy with rules. (b) Soft gating strategy with router.

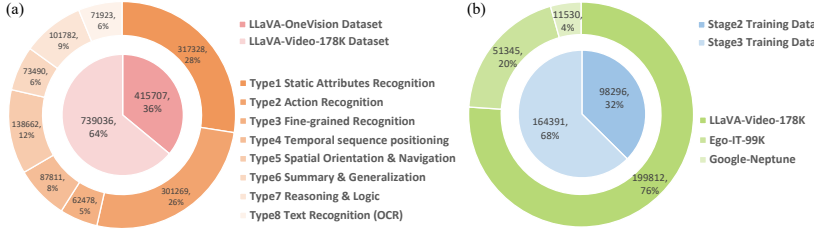


Figure 4. Dataset Composition. (a) In stage 1, the inner chart illustrates data source distribution, while the outer chart depicts 8-class distribution. (b) In stages 2-3, the inner chart compares train data allocation between consecutive stages, and the outer chart demonstrates relative contribution proportions from different data sources.

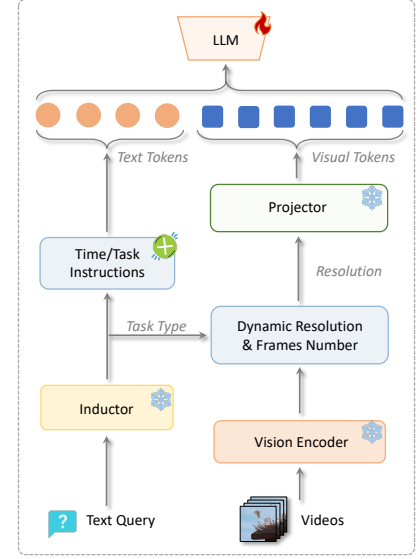


Figure 5. Training Stage 2. We first adapt the LLM to DFR, then jointly train all components in Stage 3.

is trained in Stage 1 and kept frozen during LVLM training. The MoE initialization is performed through Co-Upcycling: we duplicate each MLP layer in the Stage-2 VE four times and append randomly initialized gating networks. Thus, the MoE-VE inherits SigLIP’s pretrained contrastive visual representations rather than being trained from scratch. Meanwhile, the Projector is also replicated 4 times and configured to process four distinct input resolutions separately. This implementation results in a Soft-Gating architecture for the VE and a Hard-Gating mechanism for the Projector. For the Soft-Gated VE, we incorporate an auxiliary loss function to ensure a balanced workload distribution, with technical details presented in Sec. 4.1. Fig. 4(b) shows the Stage 2–3 data mix, with Stage 3 contributing 68%.

4. Experiments

We train TallVA with a mixture of video-instruction datasets, as demonstrated in Fig. 4, and report benchmark results, ablations, efficiency analysis, routing visualizations, and qualitative cases that inspect routing behavior.

4.1. Implementation Details

As illustrated in Fig. 4, we use essentially the same training data as the baseline, with minimal additional data to prevent overfitting; details are deferred to the appendix.

Following the baseline, we use SigLIP-400M as the VE, a two-layer MLP as the Projector, and Qwen2-7B as the LLM. The appendix reports full settings and hyperparameters. The learning rate follows:

$$LR = LR_0 \cdot \sqrt{BS/BS_0} \quad (3)$$

where BS is the global batch size, $BS_0 = 32$ is the reference global batch size, and we set LR_0 to $2.5e-5$ for LLM and $5e-6$ for VE. To achieve load balance in MoE-VE, we use auxiliary losses (Zoph et al., 2022), where $\mathcal{L}_{ori}$ is the next-token prediction loss, $\alpha_b = 1e-3$, $\alpha_z = 5e-4$:

$$\mathcal{L} = \mathcal{L}_{ori} + \alpha_b \mathcal{L}_b + \alpha_z \mathcal{L}_z \quad (4)$$

4.2. Main Results

Comparison with Previous SOTA LVLMs Using Comparable LLMs. In Tab. 1, we compare TallVA with previous SOTA models, and also present a radar chart (Fig. 6) to visualize its strengths. We evaluate on 10 challenging benchmarks spanning General Video Understanding, Temporal Reasoning, and Long Video Understanding (Fu et al., 2025; Li et al., 2024b; Mangalam et al., 2023; Xiao et al., 2021; Yu et al., 2019; Zhou et al., 2025; Wu et al., 2024a; Liu et al., 2024d). Most benchmarks are closed-ended QA with fixed answer sets; for the open-ended ActivityNet-QA benchmark, we use GPT-3.5-turbo-1106 as the judge and average multiple runs following the baseline. EgoSchema fine-tuning is isolated in Tab. 5(a), where the baseline uses the same 8% EgoIT-99K data for a fair data-efficiency comparison. Overall, TallVA achieves the best result on most benchmarks among models that employ a comparable LLM.

Comparison with LVLMs Using Stronger LLMs. We also compare TallVA against recent strong video understanding baselines presented in the gray-shaded section of

Table 1. Comparisons between TallVA and other SOTA LVLMs on competitive benchmarks. Within the non-gray comparable-LLM block, **boldface** indicates the highest score, and underlined scores denote the second-highest. TallVA scores are averaged over at least 3 runs; compared-model scores follow the cited reports. Models marked with † employ more powerful LLM backbones; we gray them out to ensure fair comparison. Scores with * indicate that TallVA outperforms at least one reported stronger-LLM baseline on that benchmark.

Model-Params	General Video Understanding					Temporal Reasoning		Long Video Understanding		
	ActivityNet-QA	MVBench	Video-MME w/o sub	EgoSchema test	PerceptionTest	NeXT-QA	TempCompass	LongVideoBench	MLVU val	MLVU test
NVILA-8B †	60.9	68.1	64.1	54.3	65.4	82.2	69.7	57.7	-	49.5
LLaVA-OneVision-72B †	60.8	56.7	66.2	62.0	66.9	79.4	67.1	63.2	66.4	47.2
VideoLLama2.1-7B	53.0	57.3	54.9	53.1	54.9	75.6	56.8	-	57.4	32.7
Qwen2-VL-7B	57.4	66.0	63.2	<u>66.7</u>	62.3	81.2	65.8	55.6	69.4	-
InternVL2-8B	56.3	63.1	58.6	-	63.5	82.5	66.0	54.6	64.0	38.4
LLaVA-OneVision-7B	56.6	56.7	58.2	60.1	57.1	79.4	64.2	56.3	64.7	<u>46.9</u>
LLaVA-Video-7B	56.5	58.6	<u>63.3</u>	57.3	<u>67.9</u>	<u>83.2</u>	65.4	<u>58.2</u>	70.8	44.8
MiniCPM-V-2.5-8B	56.1	62.3	60.9	64.5	64.4	80.3	<u>66.2</u>	56.1	66.8	41.7
TallVA-8B-A7B(Ours)	61.3*	64.5*	65.6*	75.9*	69.1*	84.0*	68.8*	59.6*	68.5*	53.4*

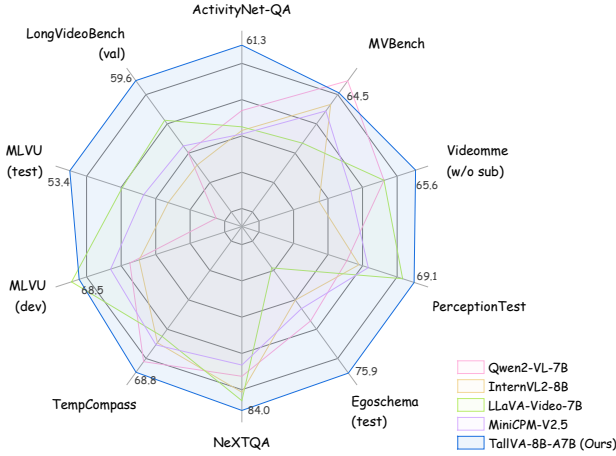


Figure 6. Radar diagram of evaluation scores. TallVA surpasses previous SOTA models in most benchmarks.

Tab. 1. These models employ much stronger LLMs compared to Qwen2-7B, with some even utilizing extensive private datasets. The asterisked scores in Tab. 1 show where TallVA exceeds at least one reported stronger-LLM baseline, supporting the effectiveness of the Task-Aware design in constructing our hybrid MoE-VT framework.

Synergy of MoE and DFR. A natural question is whether improvements stem from MoE capacity or DFR. In Tab. 3(e), all rows use the same SigLIP-400M VE and Qwen2-7B LLM, and differ only in the indicated component. MoE-ViT alone *decreases* performance (-2.7%), while DFR alone yields modest gains ($+2.8\%$). Together, MoE and DFR give the strongest tested setting ($+12.1\%$); Avg Change denotes the mean relative change over the listed benchmarks. These results suggest that DFR is a key condition for effective expert specialization in our setting. The no-DFR isolation study later in Tab. 6 further supports this interpretation.

Table 2. Inductor performance and efficiency of TallVA.

Model - Param	Acc	Time
SmolLM - 135M	86.5%	61.0 s
SmolLM2 - 135M	91.7%	55.9 s
SmolLM2 - 360M	93.5%	78.4 s
Qwen2.5 - 0.5B	92.6%	101.2 s

(a) Performance of different base models for the Inductor. Acc is measured on the test set; time is measured on 100 samples.

Model	LLaVA-Video-7B	Qwen2-VL-7B	TallVA-8B-A7B
FLOPs	$1.0844 * 10^{17}$	$1.1732 * 10^{17}$	$1.1976 * 10^{17}$

(b) End-to-end FLOPs under the standard evaluation setting, where TallVA remains comparable to Qwen2-7B LVLMs.

$top-k$ in 4	k=1	k=2	k=4
FLOPs	$1.1519 * 10^{17}$	$1.1976 * 10^{17}$	$1.2603 * 10^{17}$

(c) End-to-end FLOPs with different numbers of activated experts.

4.3. Training and Gating Ablations

Ablation Study in Stage 1. In Stage 1, we trained the Inductor model, which serves as the core module of TAM for video duration-aware question classification. The Inductor receives both video duration information and user queries to determine question categories. Starting from the base model, we incrementally incorporated two key components: Dynamic Frame Number and Resolution (hereafter referred to as DFR), along with Time/Task Instruction. The systematic ablations in Tab. 3(a) demonstrate the effectiveness of these enhancements.

Training-Priority Ablation in Stage 2. As described in Sec. 3.3, we first train the LLM in Stage 2 (Fig. 5). Since prior LVLm paradigms usually train the Projector or the entire VT first (Li et al., 2025; Liu et al., 2023; 2024b;c), this finding surprised us. While training the Projector in Stage 2 was expected to enhance performance, we observed signifi-

Table 3. Ablation studies for different training stages and components of TallVA across representative benchmarks.

	MVBench	Video-MME	MLVU-Test	
<i>Baseline (uniform budget)</i>	58.6	62.3	44.8	
<i>+ Dynamic Frame Number & Resolution</i>	59.3	63.2	45.2	
<i>+ Time/Task Instruction in Prompt</i>	59.2	63.5	44.9	
<i>+ combination of the above two</i>	59.5	63.5	46.1	
(a) Study for different routing inputs before LVLM training.				
	MVBench	Video-MME	MLVU-Test	
<i>TallVA after stage 1</i>	59.5	63.5	46.1	
<i>+ train LLM only</i>	61.8	64.4	48.5	
<i>+ train Projector only</i>	57.9	62.8	42.2	
<i>+ train VE and Projector</i>	48.3	59.1	32.7	
(b) Training-priority ablation in Stage 2.				
	MVBench	Video-MME	MLVU-Test	
<i>TallVA after stage 2 (single Projector)</i>	61.8	64.4	48.5	
<i>+ soft MoE-VE (top2 in 4)</i>	63.9	65.1	50.7	
<i>+ hard MoE-VE, based on task type</i>	56.4	58.7	35.6	
<i>+ soft MoE-VE (top2 in 8)</i>	63.2	63.8	50.0	
(c) Routing Strategies for Vision Encoder.				
	MVBench	Video-MME	MLVU-Test	
<i>TallVA after stage 2 (single VE)</i>	61.8	64.4	48.5	
<i>+ soft MoE-Projector (top2 in 4)</i>	61.9	64.0	48.5	
<i>+ soft MoE-Projector (top2 in 8)</i>	61.6	63.8	48.2	
<i>+ hard MoE-Projector, based on resolution</i>	62.6	64.7	49.3	
(d) Routing Strategies for Projector.				
	ActivityNet-QA	MVBench	MLVU-Test	Avg Change
<i>Baseline</i>	56.5	58.6	44.8	+0%
<i>+ soft MoE-VE (top2 in 4)</i>	55.4	56.7	43.5	-2.7%
<i>+ Dynamic Frame Number & Resolution (DFR)</i>	58.6	59.5	46.1	+2.8%
<i>+ MoE / DFR (TallVA)</i>	61.3	64.5	53.4	+12.1%

(e) MoE and DFR together produce stronger gains than either component individually.

cant performance degradation instead. Based on Tab. 3(b), we argue that the LLM was most strongly affected by DFR, since it had previously only handled fixed frame numbers and resolutions. The appendix provides the corresponding loss curves.

Study on Gating Strategies for VE and Projector. We demonstrate that in models like TAM, the VE achieves better performance improvements when employing Soft-Gating. In contrast, implementing Hard-Gating based on task-type negatively impacts model stability, leading to performance degradation. Conversely, for the Projector module with only a limited number of layers, resolution-based hard routing is more effective than Soft-Gating. Experimental results presented in Tab. 3(c) and Tab. 3(d) validate the effectiveness of our Hybrid Gating Strategy. Additional appendix experiments on expert swapping and routing scrambling show that MoE experts are not interchangeable.

Table 4. The robustness analysis of the Inductor module.

Confidence	0-0.2	0.2-0.4	0.4-0.6	0.6-0.8	0.8-1.0	All
Inductor Acc	0.59	0.73	0.84	0.93	0.98	92.3%
TAM Acc	0.39	0.47	0.61	0.70	0.75	68.2%
Baseline Acc	0.36	0.45	0.56	0.63	0.67	63.1%
Sample Numbers	136	544	1679	5212	2429	10000
Percentage	1.4%	5.4%	16.8%	52.1%	24.3%	100%

(a) Calibration analysis on 10k held-out samples. Inductor Acc is task-label accuracy; TAM/Baseline Acc are downstream QA accuracies.

Methods	Original Data	Paraphrase@DS	Paraphrase@GPT
Acc@Inductor	91.4%	90.2%	90.5%

(b) Paraphrasing robustness. Minimal accuracy drop confirms generalization beyond specific phrasings.

Pred. \ GT	1	2	3	4	5	6	7	8
1	92.5%	3%	3.5%	0	1%	0	0	0
2	2.5%	87.5%	3%	3%	0.5%	0	2.5%	1%
3	1.5%	3%	86.5%	1.5%	2%	0	4.5%	1%
4	0	1.5%	2%	94%	0	2%	0.5%	0
5	0	0.5%	1%	0	91.5%	0	4%	3%
6	0	1%	0	0.5%	0.5%	97%	0	1%
7	3%	3.5%	4%	1%	1%	1%	85.5%	1%
8	0.5%	0	0	0	3.5%	0	3%	93%

(c) Confusion matrix over 200 samples per task type; diagonal accuracies range from 85.5% to 97.0%.

4.4. Inductor Robustness

Inductor Selection and Calibration. We selected four text-only pre-trained mini models as Inductor candidates (Tab. 2(a)). SmolLM2-135M demonstrated good efficiency while maintaining competitive accuracy. To assess potential cascading errors from misclassification, we evaluated calibration on 10k held-out samples. In Tab. 4(a), Inductor Acc denotes task-label classification accuracy, while TAM Acc and Baseline Acc denote downstream QA accuracy bucketed by Inductor confidence. The Inductor reaches 92% overall accuracy and 59% in the 0-0.2 confidence bucket.

Confusion Matrix and ECE. The confusion matrix further characterizes where Inductor errors occur. We provide a detailed 8×8 confusion matrix across all task types in Tab. 4, with diagonal accuracies ranging from 85.5% to 97.0%. We also compute the Expected Calibration Error (ECE):

$$\text{ECE} = \sum_i p_i \cdot |\text{Acc}_i - \text{Conf}_i| \quad (5)$$

where p_i is the proportion of samples in bucket i , and Conf_i is the average confidence. The ECE is 0.226, primarily

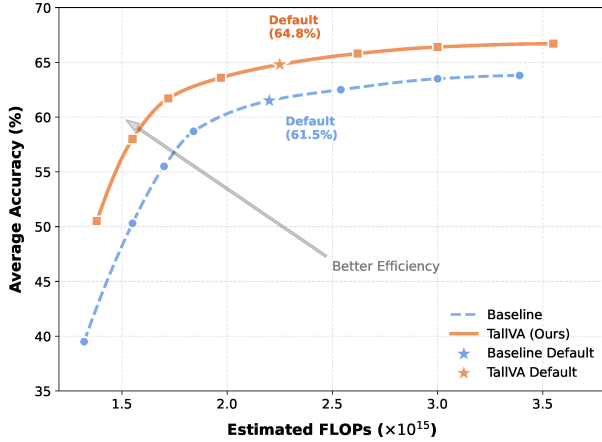
Table 5. Scalability, data efficiency, and accuracy–efficiency analysis across complementary settings.

Method	EgoSchema (before)	EgoSchema (after)	Growth Rate
TallVA (8% EgoIT-99K)	57.3	75.9	32.5%
Baseline + 8% EgoIT-99K	57.3	71.4	24.6%
EgoGPT (15% EgoIT-99K)	60.1	73.2	21.8%

(a) Data efficiency on EgoSchema.

Model	Video-MME	NExT-QA	LongVideoBench
LLaVA-Video-7B	63.3	83.2	58.2
LLaVA-Video-7B + TAM	64.1 (+0.8)	83.5 (+0.3)	58.7 (+0.5)
LLaVA-Video-72B	70.5	85.4	61.9
LLaVA-Video-72B + TAM	71.9 (+1.4)	85.5 (+0.1)	63.1 (+1.2)

(b) Scalability to 72B LLMs via LoRA fine-tuning.



(c) Visual-processing Acc–FLOPs tradeoff. FLOPs in this panel exclude the LLM.

driven by *under-confidence*, which is less harmful than over-confident misrouting. TAM consistently outperforms the baseline across all confidence buckets.

Fallback Strategies. Though the Inductor already has high accuracy, we still use a conservative fallback policy: the deterministic scheduler in Fig. 2(b) applies duration-dependent sampling intervals (0.5s/1s/2s), a 120-frame cap, and coarser pooled grids when a high-resolution choice would exceed the visual-token budget. Explicit video metadata is always provided to the LLM. This bounds the visual budget for very short or long videos even when task-regime prediction is imperfect, helping stabilize edge cases. Representative budget examples are provided in the appendix.

Paraphrasing Experiments. We conducted experiments on the Inductor with 1k queries paraphrased by DeepSeek-V3 and GPT-4o (Tab. 4(b)). The small drop (about -1%) indicates robust Inductor generalization.

4.5. Generalization and Data Efficiency

Scalability to Larger Models. Applying TAM via LoRA to LLaVA-Video-7B and 72B improves all reported metrics (Tab. 5(b)), with larger 72B gains on Video-MME (+1.4%)

and LongVideoBench (+1.2%). Appendix experiments further show that defining a “very-long-video” task type with ~800 samples improves long-video metrics with a short-video trade-off on NExT-QA, indicating low-cost but targeted extensibility.

Data Efficiency. With only 8% data of EgoIT-99K, TallVA improves EgoSchema by 32.5%, exceeding the baseline (24.6%) and EgoGPT with 15% data (21.8%; Tab. 5(a)). On EgoSchema, this relative gain suggests that task-aware visual budgeting can improve evidence allocation in a low-data regime, rather than relying only on broader benchmark-specific fine-tuning.

4.6. Specialization and Efficiency Analysis

Expert Specialization and Efficiency. In Fig. 7, we record average routing weights across MoE-VE layers and duration-conditioned task groups, showing that video length changes expert allocation. TallVA’s extra parameters mainly come from upcycling the FFN parameters of SigLIP-400M; under the standard evaluation setting, its end-to-end FLOPs remain comparable to dense 7B video LVLMs (Tab. 2(b,c)). Separately, Tab. 5(c) reports visual-processing FLOPs only (excluding the LLM) and shows better accuracy at comparable visual compute.

A routing-granularity ablation compares uniform budgeting, a 3-class coarsening, and the full 8-class routing abstraction. The 3-class variant is already helpful, while full TAM remains stronger, showing that the eight labels preserve visual-budget distinctions inside coarse families.

Complementarily, Tab. 6 keeps the LLM and projector fixed and removes query-aware frame/resolution budgeting. All sparse MoE configurations underperform the dense baseline, indicating that DFR is a key condition for useful expert specialization. Taken together, these ablations clarify that TAM’s task labels are visual-budget regimes: some tasks are mainly frame-sensitive, some are resolution-sensitive, and some require a balanced allocation.

Without DFR, sparse visual experts appear to receive mixed visual inputs, making reliable specialization to the query-relevant evidence type difficult. With DFR, the input budget is structured before visual encoding, so MoE-VE can specialize over coherent frame/resolution regimes while the hard-gated Projector handles pooled-resolution differences. Additional appendix analyses of expert swapping and routing scrambling support the same interpretation.

Robustness and Scope. TAM does not require perfect task classification: for ambiguous queries, the Inductor predicts a dominant regime while the fallback policy keeps duration-aware frame and pooled-grid bounds. The paraphrasing and calibration results in Tab. 4 show that this conservative routing remains robust under wording changes

Table 6. Impact of MoE-ViT Isolation (without DFR). We evaluated MoE-ViT configurations while keeping the LLM and projector fixed. The results indicate that simply replacing the vision encoder with an MoE-ViT (without query-aware routing) leads to performance degradation across benchmarks.

Model	Baseline	Top2 in 8	Top2 in 4	Top1 in 4
Video-MME	63.3	61.5	62.8	63.0
MVBench	58.6	53.3	56.7	56.5
MLVU-Test	44.8	42.0	43.5	42.2
Relative Avg Change	+0%	−5.98%	−2.21%	−3.01%

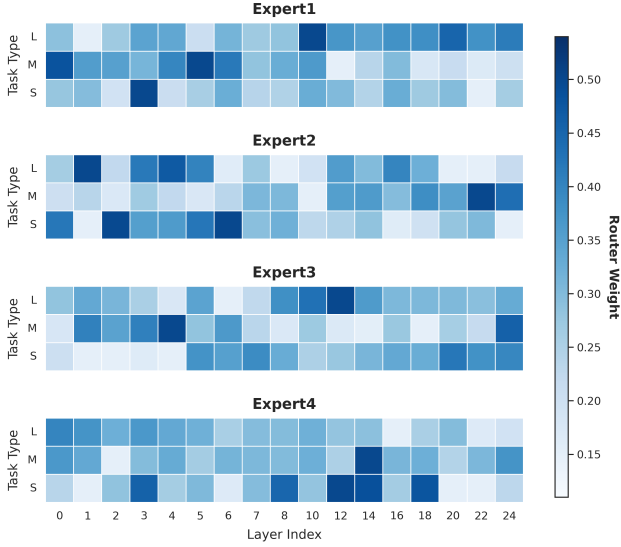


Figure 7. Visualization analysis for MoE-VE across layers and duration-conditioned task groups.

and low-confidence predictions.

These experiments also set the boundary of the current design. TAM does not claim to choose globally optimal frames or replace content-aware key-frame discovery or token pruning; it first chooses a task-level frame and resolution budget before visual encoding. The remaining question of which frames or patches to keep inside that budget is therefore orthogonal to TAM’s scheduler, as discussed in Sec. 5.1.

5. Discussion

5.1. TAM as a Composable Budget Layer

TAM is a task-level visual-budget layer, not a replacement for content-level evidence selection. Its scheduler decides how much temporal and spatial evidence a query should receive before visual encoding, whereas key-frame selection, saliency scoring, and token pruning decide which frames or patches to keep inside that budget. The two decisions are therefore complementary: TAM can provide broad temporal coverage for summary queries or higher spatial detail for OCR-like queries, while later selectors or compressors remove redundancy within the chosen budget. This also

gives a practical deployment path: systems can log and tune TAM’s task regime, frame count, pooled token grid, and projector route, while downstream modules diagnose whether failures stem from budgeting or evidence selection.

5.2. Conclusion

In this paper, we introduce Task-Aware Mechanism (TAM), a hybrid-gated MoE Vision Tower framework that adapts frame count and resolution according to the user query and video length. TAM uses an 8-class routing abstraction, TA-116K supervision, and a compact 0.1B Inductor to convert each query into an explicit visual-computation budget before visual encoding. With these components, TallVA achieves the best result on most benchmarks among comparable-LLM models and remains competitive with stronger-LLM systems.

- A key finding is that MoE-ViT and DFR work best together: applying them separately gives limited gains, while their combination is strongest (+12.1%), suggesting that sparse experts specialize effectively only with task-appropriate visual inputs.
- The 0.1B Inductor remains robust under paraphrased and low-confidence queries and is extensible to new task types; the scheduler’s duration-aware fallback also keeps visual budgets bounded when task-regime prediction is imperfect.
- TAM is compatible with stronger LLM backbones, low-data adaptation, and downstream frame-level selection or token compression, and it improves the visual-processing accuracy–compute tradeoff in our ablations.

More broadly, TAM keeps the contribution focused on interpretable, controllable visual budgeting rather than replacing content-level evidence selection. We hope this direction encourages follow-up work on low-cost task-aware routing for long-video LVLs and on visual pipelines that allocate computation according to both the question being asked and the video being observed.

Impact Statement

This work improves task-aware video understanding in multimodal expert models. Potential benefits include accessibility, education, and assistive search or summarization of long videos. The same capabilities may lower the cost of large-scale video analysis in sensitive settings such as monitoring, moderation, or profiling. If used improperly, such systems may harm privacy, amplify biased or unreliable interpretations of video evidence, or support intrusive surveillance and automated decision-making. TAM exposes its selected visual budget, so deployments should audit these choices and pair them with consent-aware data governance and human review in consequential settings.

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A. More Details About Training Data Collecting

A.1. Tagging Prompts

To train the Inductor to classify different tasks, we selected 116K high-quality, non-redundant queries (excluding answers) from open-source datasets. Subsequently, we annotated these queries using the DeepSeek-R1 model, ultimately forming the TA-116K dataset. For clarity, we present the prompts in tabular form. The content of our prompts is as follows.

Generally, the video understanding tasks can be categorized into the following eight types:

Table 7. Prompts and descriptions used for tagging and describing our classification strategy to DeepSeek-R1.

Category	Description	Example
Static Attributes Recognition	Identifying perceptible static attributes of people/objects (color, quantity, state, material, etc.), or existence queries. Includes object retrieval and counting problems.	- What color is the helicopter? - How many riders are on the podium? - How many people are there in the video?
Action Recognition	Recognizing dynamic events/actions and interaction relationships (excluding facial expressions). Includes methods of achieving actions.	- What does the person do after opening a can of chickpeas? - How does the skier move down the slope? - How did the man make his way to his home?
Fine-grained Attributes Recognition	Identifying advanced attributes like poses, emotions, and facial expressions, typically requiring subtitle/dialogue analysis.	- Does the man feel sorry for the woman? - Is the woman happy or angry at the end of the video?
Temporal Sequence Positioning	Determining event sequences/timings, causal order, or occurrence counts (limited to video content).	- Arrange the events in chronological order. - What happened after the man left the house? - How many times did the brightness change?
Spatial Orientation and Navigation	Recognizing positions/spatial relationships or navigation sequences in environments.	- Where is the yellow container located? - Where is the kitchen, next to the bedroom or next to the washroom?
Summary and Generalization	Extracting narrative structures, themes, or actionable insights through content summarization.	- Analyze the video's narrative structure. - What's the video mainly about? - According to the video, how can we solve the problem?
Reasoning and Logic	Inferring causes, motivations, purposes, or relationships between entities based on video content.	- Why did the toddler cry at the end? - What indicates that the food is being cooked? - What made the man angry? - Is their relationship more like friend or enemy?
OCR and Cross-modal Alignment	Cross-modal alignment between video content and text (e.g., subtitle matching), or identifying on-screen text.	- What objects appear after the subtitle mentions 'multiple reasons'? - Match the dialogue to the scene.

You can take answer options into account when available. Ignore non-semantic content (e.g., "Answer using short words") when analyzing questions.

A.2. Classification Strategy

As shown in Tab. 7, we divide video understanding tasks into eight types. Our main classification criteria are the inherent meaning of the tasks and their different sensitivities to frame count and resolution. When considering only the sensitivity to these two factors, the tasks can be categorized into the following three groups:

1. Tasks more sensitive to resolution: tasks that depend on fine details, such as Static Attributes Recognition and Spatial Orientation and Navigation, are sensitive to resolution. MoE-VT will provide inputs with fewer frames but higher resolution.
2. Tasks more sensitive to frame count: tasks like Temporal Sequence Positioning and Summary and Generalization become significantly more complex in long videos. MoE-VT will provide inputs with more frames but lower resolution.
3. Balanced tasks: some tasks may span multiple categories. For example, a Reasoning and Logic task might involve aspects of Action Recognition along with logical reasoning. MoE-VT will provide inputs with a balanced number of frames and resolution.

Counting and retrieval-style questions can fall into different groups depending on what evidence is required. For instance, counting static objects in a single scene is resolution-sensitive, while counting repeated actions over time is frame-count-sensitive. The Inductor therefore predicts the dominant visual-budget regime rather than a mutually exclusive semantic type.

We additionally verified TA-116K label quality through human audit. Three independent groups each re-labeled 400 samples (50 per category), achieving 95.5%, 93.0%, and 94.0% agreement with the DeepSeek-R1 labels, respectively.

Table 8. Representative DFR scheduler outputs. Entries show pooled token grid and frame count under a 20.5K visual-token budget and 120-frame cap. High-resolution 27×27 cases are capped at 28 frames, while 11×11 is selected when preserving long-video temporal coverage would otherwise exceed the budget. Exact frame count can further shrink when text tokens consume more context.

Task Type	15s Video	60s Video	300s Video
Static Attributes Recognition	27×27 , 28f	27×27 , 28f	14×14 , 104f
Action Recognition	18×18 , 30f	18×18 , 60f	11×11 , 120f
Fine-grained Attributes Recognition	27×27 , 28f	27×27 , 28f	18×18 , 60f
Temporal Sequence Positioning	14×14 , 30f	14×14 , 60f	11×11 , 120f
Spatial Orientation and Navigation	27×27 , 28f	27×27 , 28f	18×18 , 60f
Summary and Generalization	18×18 , 30f	18×18 , 60f	11×11 , 120f
Reasoning and Logic	27×27 , 28f	27×27 , 28f	18×18 , 60f
OCR and Cross-modal Alignment	27×27 , 28f	18×18 , 60f	18×18 , 60f

B. Dataset Composition

In the main text of the paper, we have already presented the composition and proportion of the training data used in Stage 2 and Stage 3. To reduce the chance that gains come mainly from extra data, we used almost the same data as in the baseline for training. To prevent overfitting, we also use a subset of samples from the EgoIT-99K(Yang et al., 2025) and Neptune(Nagrani et al., 2024) datasets, which make up no more than a quarter of our training dataset.

In the code repository released with the final artifacts, readers will also find the specific data proportions for these stages stored in files such as `./scripts/TallVA/all_data_stagex.yaml`.

C. Additional Representative Case Studies

As referenced in the main text, Fig. 8 demonstrates the performance of TallVA on two distinct styles of tasks and compares it with a baseline model, showcasing strong adaptability across different tasks. To further highlight the comprehensive capabilities of TallVA, we provide additional case studies here. For consistency, we highlight **correct content** and **incorrect content**

with different colors. We also visualize the output of Inductor and MoE-VT. However, it should be noted that the Inductor outputs only the task type, while the figures show the Task Instructions added to the LLM.

These cases cover four complementary settings: short videos that focus on fine details in Fig. 11; the challenging dialogue examples from the main text in Fig. 8; videos with completely different styles, including 3D animations that were not present in the training data, in Fig. 9; and deeper understanding of plots and content, involving more human-like activities such as match commentary or watching a funny video, in Fig. 10.

Together, these examples serve as qualitative diagnostics rather than standalone quantitative evidence. They show the intended routing behavior: short detail-focused questions receive higher-resolution visual inputs, while long narrative or temporal questions receive broader frame coverage.

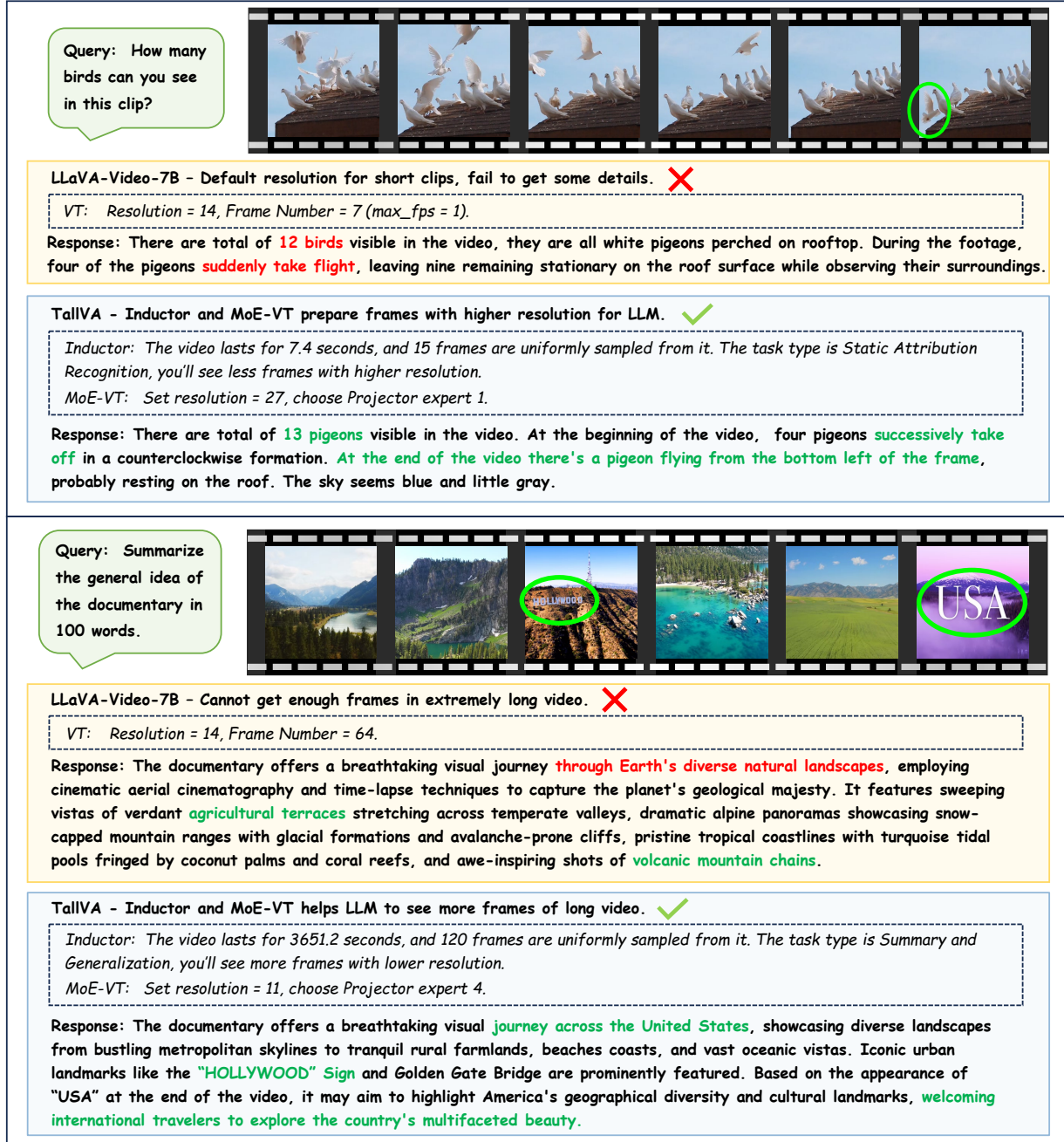


Figure 8. Dialogues between User and Model on Challenging Tasks. We highlight correct and incorrect content. We also visualize the outputs of Inductor and MoE-VT. Additional cases are shown throughout this appendix.



Figure 9. **More Case Studies on Different Video Styles.** We demonstrate realistic ultra-long videos as well as 3D animations. TallVA shows better performance across different video styles.

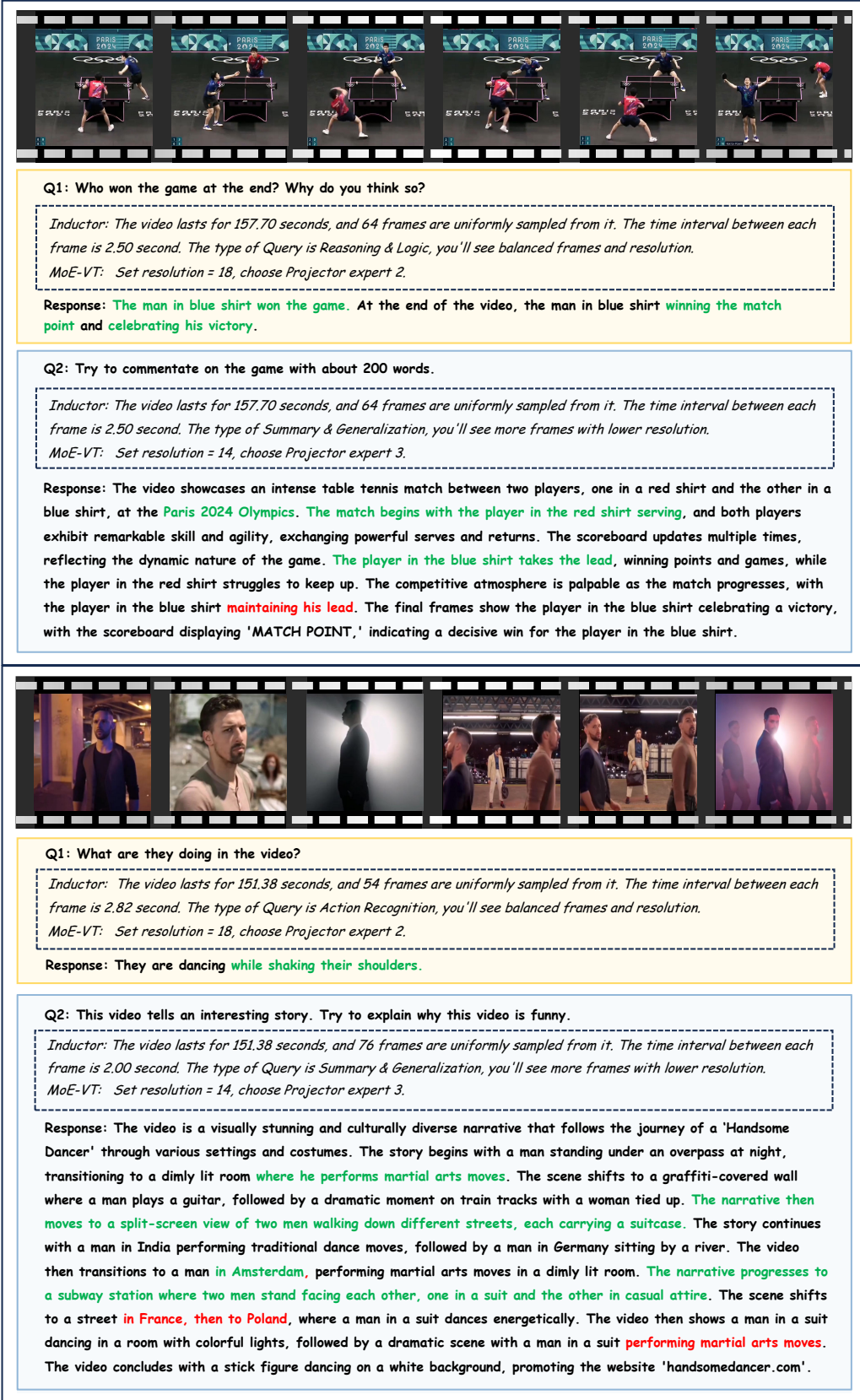


Figure 10. Case Studies for Deeper Understanding of Plot and Content. TallVA may hallucinate on high-difficulty problems. Future work can use more complete datasets and stronger LLMs to reduce this issue.

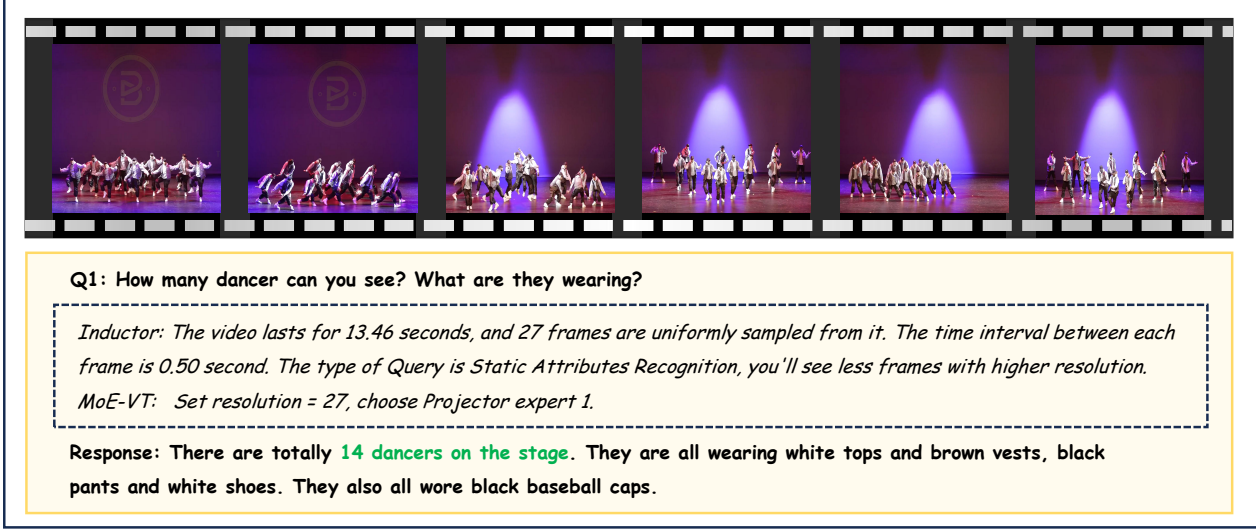


Figure 11. **Case Study on Short Clips.** For shorter videos, TallVA can use higher-resolution frames to help understand detailed video content.

D. Discussion: Future Directions and Design Considerations

In this section, we discuss design questions that arose during the development of TAM: the relationship between task-aware routing and difficulty estimation, the token-count details behind TAM’s scheduler, and the role of hard-coded routing in future LVLm design. The broader composability argument is now presented in Sec. 5.1.

D.1. TAM and Difficulty-Aware Routing

A natural question is whether TAM should incorporate sample-wise difficulty estimation to allocate visual budgets more precisely. We carefully considered this direction and concluded that, while intuitive, difficulty-based routing faces fundamental challenges for video understanding.

The core problem is that understanding “what makes a question hard” is closely related to answering it. Existing difficulty estimation methods mainly fall into two categories, neither of which works well for our setting. Model cascade approaches—running a weak model first and marking failures as “hard”—require multiple forward passes through large models, exactly what we want to avoid with a lightweight Inductor. Heuristic approaches like modeling frame-to-frame entropy completely ignore the user query; yet the same video can be trivial (“What color is the car?”) or extremely hard (“Count all people in the background”) depending on what is asked.

Even if we had a reliable difficulty score, translating it into frame count and resolution decisions is non-trivial. Consider that for a 2-hour movie, “What is the weather like?” is easy—a few frames suffice—while for a 10-second clip, “How many plants are on the lawn?” might be very hard, requiring high-resolution frames and careful counting. Moreover, a “hard” OCR question needs higher resolution, whereas a “hard” temporal reasoning question needs more frames. A single difficulty score cannot tell us *where* to spend the budget. This is fundamentally different from text-only tasks, where we can often improve performance on hard questions by giving the model more “thinking time.” For video understanding, if we missed the key frame, no amount of reasoning will recover the lost information.

Given these challenges, TAM takes a pragmatic approach: rather than chasing sample-wise difficulty, we classify queries into task types that reflect different patterns of frame-count and resolution requirements. Within each type, the actual budget still varies with video length (see the case studies in Appendix C). Our experiments demonstrate that this simple task-aware and length-aware scheme already yields clear gains.

That said, we do see a role for difficulty estimation in future work. The key insight is that difficulty might be better used to allocate *reasoning* budgets rather than *visual* budgets. One could imagine a system where TAM handles the visual side, and a separate module decides how much chain-of-thought or iterative refinement to apply based on query difficulty.

Reasoning-centric multimodal models remain underexplored, and TAM may serve as a foundation that could later be combined with difficulty-aware reasoning modules.

D.2. Token-Count Control Details

Sec. 5.1 discusses how TAM can be stacked with frame-selection and token-compression methods. Here, we only spell out how TAM controls token count before such content-level modules operate. Taking SigLIP-so400m-patch14 as an example, each frame is divided into a 27×27 patch grid, which is then pooled via bilinear interpolation before projection. By changing the target pooled grid, TAM controls how many visual tokens each frame contributes: $27 \times 27 = 729$ tokens for the highest resolution, $18 \times 18 = 324$ tokens, $14 \times 14 = 196$ tokens (default), and $11 \times 11 = 121$ tokens for the lowest resolution. Thus, TAM controls token count through both the number of frames sampled before the VE and the pooled spatial density retained before the Projector.

This differs from pruning or merging inside a fixed-resolution input: TAM decides the task-level budget first, while later methods can still remove redundant frames or patches within that budget.

D.3. The Role of Hard-Coded Routing

One might view TAM’s use of predefined routing rules as a limitation—should we not learn everything end-to-end? We argue that hard-coded routing is actually a reasonable design choice for the current stage of research, and we explain our reasoning below.

First, we emphasize that TAM is not purely hard-coded but employs a *hybrid* gating strategy. The MoE-ViT uses soft routing where experts are selected based on learned gates. The MoE-Projector uses hard routing based on resolution. Frame count and resolution are determined by task type and video length via predefined rules. Our ablations (Table 3 in the main paper) show that this combination works better than using soft or hard routing uniformly—different components benefit from different strategies.

Why not learn frame count and resolution end-to-end? We considered this carefully, but there are practical obstacles. Current ViT architectures like SigLIP and CLIP are not designed for variable-resolution inputs in a differentiable way. We would need massive amounts of data annotating “the right frame count and resolution for this query,” which does not exist. Furthermore, frame sampling is inherently discrete—you cannot sample 16.5 frames—making gradient-based learning difficult. Hard-coded rules allow us to validate the core idea that task-aware visual budgeting helps, without needing to solve all these challenges simultaneously.

For practical deployment, hard-coded routing offers several advantages. The Inductor can be trained independently and frozen during LVLM training. Routing rules can be quickly tested and adjusted without retraining the whole model. Resource usage is predictable, which is important for edge devices with limited memory. The system’s behavior is also interpretable: developers can understand why a particular frame count was chosen for a given query.

We view this work as a stepping stone rather than the final answer. The research trajectory we envision proceeds as follows: first, validate that task-aware budgeting helps using simple rules (this work); then, explore learned routers that can optimize frame and resolution selection jointly; eventually, develop fully differentiable pipelines that can adapt to new task types without manual intervention. TAM establishes that the *goal*—adapting visual processing to the task—is worthwhile. The specific *mechanism* (hard vs. soft routing) is a design choice that can evolve as the field matures.

D.4. Broader Impacts

Our approach enhances the ability of LVLMs to understand video content by providing task-appropriate visual inputs, contributing to more capable multimodal AI systems. Publicly released code and weights, when available, can accelerate community development, though they may also carry potential risks. We will include appropriate licenses and release notes to guide responsible use of our model.

E. Training Sets and Hyperparameters

Tab. 9 shows our specific hyperparameter settings for Training Stages 2 and 3. We used the same learning rate (LR) and LR scheduler for both stages. We employed the AdamW optimizer with a batch size of 1, gradient accumulation of 2, and 16 GPUs using Zero1-offload and Zero2-offload. The Vision Encoder and Projector were frozen while the LLM was trained in Stage 2. Additionally, we incorporated MoE Blocks to enhance the model’s capacity in Stage 3. The maximum context length (including visual tokens and text tokens) is set to 21000. For Stage 1 Inductor training, we use SmolLM2-135M-Instruct for 3 epochs with learning rate 5e-5, train/eval batch sizes 48/32, bf16, AdamW, DeepSpeed ZeRO-1 on 8 GPUs, max length 8192, and a 10% held-out split. Training takes about 2.4k A100 GPU hours in total for Stages 2–3. For further details, please refer to Tab. 9.

Table 9. Hyperparameters for the second and third stages of training.

Hyperparameter	Stage 2	Stage 3
Learning Rate	2.5×10^{-5}	2.5×10^{-5}
Vision Tower LR	–	5×10^{-6}
LR schedule	Cosine	Cosine
Batchsize per GPU	1	1
Gradient Acc.	2	2
GPU Number	16×A100	16×A100
Zero	Zero1-offload	Zero2-offload
Optimizer	AdamW	AdamW
Projector	Freeze	Train
Vision Encoder	Freeze	Train
LLM	Train	Train
MoE Blocks	–	✓
Max Context Length	21000	21000
Max Output Token	2048	2048
Warm Up Ratio	0.03	0.03
Total Steps	1200	3200

F. Reproducibility Statement

In this paper, we strive to enhance the reproducibility of our work. The overall workflow is depicted in Fig. 2, and the detailed structure of the training dataset is illustrated in Fig. 4. Additionally, we provide comprehensive details of the experimental setup in Sec. 4.1, along with key hyperparameters listed in Appendix E.

To support reproducibility and further research, we have completed the organization of the TA-116K dataset and have made it publicly available at <https://huggingface.co/datasets/Giesen/TA-116k>. TA-116K serves as the training and testing dataset for the Inductor module. It is annotated by DeepSeek-R1 and is released to promote reproducibility, transparency, and future research.

To support full reproduction, we will continue releasing cleaned implementation code, model weights, and evaluation scripts with documentation as they are finalized.

G. Additional Experiments and Analysis

This section provides supplementary experiments referenced in the main paper, including routing granularity, MoE isolation studies, expert interchangeability tests, task extensibility, and accuracy–efficiency tradeoff analysis.

G.1. Routing Granularity Ablation

We compare uniform budgeting, a 3-class coarsening, and the full 8-class TAM routing abstraction. The 3-class setting merges the eight task types into high-resolution, high-frame, and balanced visual-budget families, while the full abstraction preserves finer distinctions within those families.

Table 10. Routing granularity ablation. The 3-class setting merges the 8 task types into high-resolution, high-frame, and balanced visual-budget families.

Budgeting	ActNet	MVB	VME	Ego	TempCompass	MLVU
Uniform	56.5	58.6	63.3	57.3	58.2	44.8
3-class	59.8	62.3	65.0	72.4	66.2	51.1
8-class (TAM)	61.3	64.5	65.6	75.9	68.8	53.4

G.2. MoE Isolation Study

As discussed around Tab. 6, we investigate whether performance gains stem from MoE capacity or query-aware routing. Tab. 6 in the main paper shows that simply replacing the vision encoder with an MoE-ViT (without Dynamic Frames and Resolution) leads to performance degradation across all configurations.

This result aligns with prior work (Li et al., 2024a) showing that MoE-izing the vision encoder alone does not automatically yield gains. The MoE experts require task-appropriate visual inputs (enabled by DFR) to specialize effectively.

G.3. Expert Interchangeability Tests

To verify that MoE routing is causally important, we performed expert-swap and routing-scramble experiments.

Table 11. Causal Analysis of MoE-ViT Routing. We compare standard routing against Random Top-2 and Lowest-2 selection. The significant performance drop (up to 12.5%) confirms that the learned router identifies necessary experts rather than selecting arbitrarily.

MoE-ViT Configuration	Video-MME	MVBench	Relative Avg Change
Normal TAM	65.6	64.5	+0%
Random Top-2	60.3	58.2	−9.8%
Lowest 2 Experts	56.1	55.7	−12.5%

Table 12. Causal Analysis of MoE-Projector Routing. Since the Projector uses hard-gating based on resolution, we test *Random* mapping and *Expert Swap* (swapping high-res and low-res experts). The drastic drop (−13.6%) in Expert Swap confirms that experts are highly specialized for specific resolutions.

MoE-Projector Config	Video-MME	MVBench	Relative Avg Change
Normal TAM	65.6	64.5	+0%
+Random Experts	63.2	62.7	−3.2%
+Expert Swap	58.3	54.1	−13.6%

For MoE-ViT (Tab. 11), Random Top-2 selection degrades performance by 9.8%, and selecting the Lowest-2 experts causes 12.5% degradation. For MoE-Projector (Tab. 12), swapping the highest-resolution and lowest-resolution experts leads to 13.6% degradation. These results confirm that both the learned routing in MoE-ViT and the resolution–expert mapping in MoE-Projector are essential—experts are *not* interchangeable.

G.4. Low-Cost Task Extensibility

To demonstrate that TAM can flexibly adapt to new requirements, we defined a new “very-long-video” task type using approximately 800 samples and fine-tuned only the 135M Inductor via LoRA (2 epochs), keeping the full LVLM frozen.

Table 13. Low-Cost Extensibility via Inductor Adaptation. We defined a new task type ("very-long-video") using ~ 800 samples and fine-tuned only the 135M Inductor via LoRA. This minimal adaptation improves long-video metrics, with a short-video trade-off on NExT-QA, demonstrating that TAM can flexibly adapt to new requirements without retraining the full LVLM.

Inductor Configuration	Video-MME (Long)	LongVideoBench (Long)	NExT-QA (Short)
Original Inductor	65.6	59.6	84.0
Inductor after LoRA Finetune	66.2	60.8	82.7

As shown in Tab. 13, this minimal adaptation yields clear gains on long-video benchmarks (Video-MME +0.6, LongVideoBench +1.2) with a slight trade-off on short videos. This confirms that new task types can be added at very low cost without retraining the full model.

G.5. Accuracy–Efficiency Tradeoff Details

As shown in Tab. 5(c) in the main paper, we evaluated the visual-processing Accuracy–FLOPs tradeoff by varying frame counts and patch resolutions on samples from LongVideoBench and MLVU. The detailed operating points are:

Table 14. Visual-processing accuracy–FLOPs tradeoff of TallVA and Baseline. These FLOPs exclude the LLM and are therefore separate from the end-to-end FLOPs reported in Tab. 2. We selected representative operating points showing that TAM has better efficiency under comparable visual compute.

<i>Baseline FLOPs</i>	1.32×10^{15}	1.55×10^{15}	1.84×10^{15} (<i>Default</i>)	2.54×10^{15}	3.39×10^{15}
<i>Baseline Acc</i>	39.5%	50.3%	61.5%	62.5%	63.8%
<i>TallVA FLOPs</i>	1.38×10^{15}	1.72×10^{15}	1.97×10^{15} (<i>Default</i>)	2.62×10^{15}	3.55×10^{15}
<i>TallVA Acc</i>	50.5%	61.7%	64.8%	65.8%	66.7%

At default settings, TallVA uses 1.97×10^{15} visual-processing FLOPs compared to the baseline’s 1.84×10^{15} FLOPs (+7%), while achieving higher accuracy (64.8% vs 61.5%). Wall-clock latency follows similar trends: 6.2s vs 5.7s per sample; peak memory is 61.2GB vs 57.5GB.

H. Loss Analysis in Training Stage 2

In the main text, we mentioned that "our extensive experiments suggest that the LLM should be trained first in Stage 2." We believe that this is effective because the application of Dynamic Frame Number and Resolution has the greatest impact on the LLM. Therefore, the LLM needs to learn how to handle varying numbers and resolutions of frames first; otherwise, errors will accumulate in the LLM, ultimately making it difficult to train the Vision Tower.

To validate our hypothesis, we present the loss curves in Fig. 12 of different training strategies in Stage 2. We selected three typical scenarios: training the LLM first, training the Projector first, or training the entire VT (Vision Encoder + Projector). As shown in Tab. 3, the loss curve of LLM-only training decreases rapidly at the beginning of training, whereas the other two approaches exhibit an upward trend in loss along with a significant drop in actual performance.

I. More Related Works

Recent long-video LVLM research has explored several directions that are complementary to TAM. Memory- and streaming-oriented models reduce long videos into compact histories or online states, including MovieChat, MA-LMM, Streaming Long Video Understanding, and VideoLLM-online (Song et al., 2024; He et al., 2024a; Qian et al., 2024; Chen et al., 2024a). Another line increases the effective visual context through long-context transfer, system-level scaling, hour-scale modeling, or adaptive spatiotemporal compression (Zhang et al., 2025b; Chen et al., 2025; Shu et al., 2025; Shen et al., 2025).

Recent adaptive methods are especially close in spirit to our task-aware view: query-guided or learned frame selection methods select informative frames under a fixed or implicit budget (Liu et al., 2025; Tang et al., 2025; Buch et al., 2025; Wang et al., 2025), while adaptive projector or inference designs route visual features according to instructions or latency constraints (Zhao et al., 2025; Xu et al., 2025). TAM differs from these efforts by making the task-level frame/resolution budget explicit before the video enters the hybrid MoE vision tower, so frame selection, token compression, and long-context

modeling can be viewed as downstream or complementary modules.

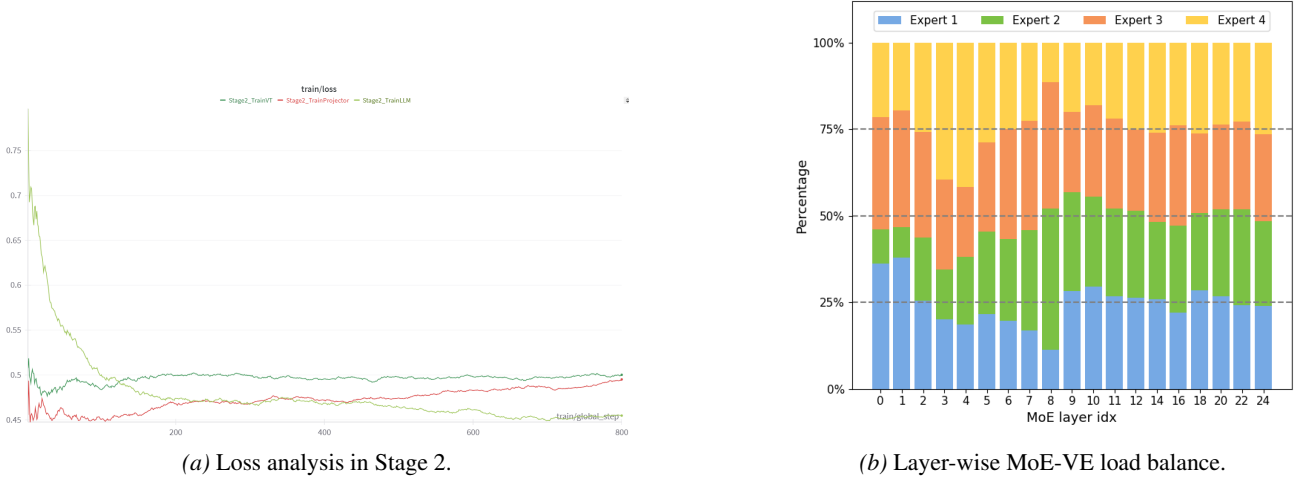


Figure 12. **Training and Visualization Analysis.** (a) Loss analysis in Stage 2; (b) MoE-VE layer-wise visualization with load balancing.

J. Limitations

Due to time and resource constraints, most full-training experiments are conducted on the 7B model. We provide preliminary LoRA transfer results on LLaVA-Video-72B in Tab. 5, but larger-scale full training and longer-context settings remain future work. Secondly, as a video understanding model, TallVA still exhibits hallucination issues, with additional successful and failed cases presented in Appendix C. In future work, we will evaluate TAM on more professional-domain benchmarks and collect more data to optimize the classification strategy. Applying TAM to models with longer context lengths (such as 128K tokens) remains a promising direction for further study.